AP PRECALCULUS

Unit 1 • Lesson 1.13-1.14 • Notes (filled in)

Topic 1.13-1.14 • Function Model Construction

Notes (filled in)

Learning goals

- Choose an appropriate function type for data or a context, and state the assumptions.
- Build models from geometric or verbal descriptions, and from data with regression.
- Use models to predict and interpret, with units and a sensible domain.

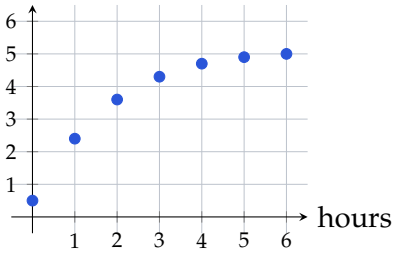
Vocabulary. model, linear, quadratic, cubic, rational, piecewise, regression, assumption, restricted domain, extrapolation.

1. Choosing a model

Key idea

Roughly constant rate of change	linear
Constant second differences; one maximum or minimum	quadratic
Two turning points or a change in concavity	cubic
Approaches a limiting value; "per unit" quantities	rational
Different rules on different intervals	piecewise

Example 1 • Which model?



The concentration of a medicine rises, then levels off near 5. Which model fits?

The data approach a limiting value: a rational model with horizontal asymptote $y = 5$ fits. A polynomial would eventually grow without bound.

2. Assumptions and domain

Every model assumes the pattern continues. Restrict the domain to inputs that make sense in context (for example, time ≥ 0, lengths positive). Predictions far outside the data (extrapolation) are less reliable.

3. Building a model from a description

Example 2 • Optimization

A farmer has 60 m of fence for a rectangular pen along a wall (no fence on the wall side). Model the area and find the largest area.

Let x = length of each side perpendicular to the wall. The parallel side is $60 - 2x$. $A(x) = x(60 - 2x)$, domain $0 < x < 30$.
$A(x) = -2(x - 15)^2 + 450$: maximum area 450 m^2 when $x = 15$ m (and the parallel side is 30 m).

Example 3 • Piecewise

A phone plan costs \$20 per month including 100 texts, plus \$0.10 per extra text. Write $C(n)$ and find $C(250)$.

$C(n) = 20$ for $0 \le n \le 100$; $C(n) = 20 + 0.10(n - 100)$ for $n > 100$. $C(250) = 35$: 250 texts cost \$35.

4. Building a model from data

Example 4 • Regression (calculator)

A ball's height (m) at $t = 0, 1, 2, 3, 4$ s is $1.2,\ 6.1,\ 9.0,\ 9.9,\ 8.6$. Find a quadratic model and predict the height at $t = 5$.

Quadratic regression: $h(t) \approx -1.029t^2 + 5.974t + 1.183$. $h(5) \approx 5.3$ m. Maximum about 9.9 m at $t \approx 2.9$ s.

AP exam tip
Keep regression coefficients stored in the calculator and round only the final answer (three decimal places). Always state units and the meaning in context.

Common mistakes

- Leaving two variables in a model: use the constraint to eliminate one.
- Using a model for inputs outside its domain (negative times, negative lengths).

Check your understanding

1. "The value drops by \$1200 each year" suggests a linear model.
2. A box with a square base of side x and volume 32 has height $\frac{32}{x^2}$.